Governance challenges and ethical risks of Agentic AI in education: accountability, hallucination, bias amplification, and control mechanisms

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***Abstract*—This article addresses the emerging governance challenges posed by agentic Artificial Intelligence (AAI) in education, where autonomous systems can plan, act, and adapt with limited human oversight. The existing AI ethics and governance frameworks, largely designed for toollike or assistive AI, do not adequately cover hallucinated decisions, bias amplification, and accountability gaps introduced by persistent, self-directed educational agents. The study's objective is to conceptualize how agentic autonomy reshapes ethical risk in education and to propose governance mechanisms that preserve human agency and institutional trust. So, how do hallucination, bias amplification, and accountability gaps manifest in AAI for education, and what governance controls can mitigate these risks while maintaining pedagogical integrity? Methodologically, the study adopts a qualitative, conceptual, and analytical design. Based on systematic literature analysis of peer-reviewed work on AAI, AI governance, and AI ethics in education (2020–2025), the findings are interpreted through RAIGT and the Human-Centered AI Framework to enable theoretical triangulation. The analysis reveals that: hallucinated decisions become more consequential as agentic systems propagate errors across autonomous action chains; feedback loops intensify bias amplification in assessment, recommendation, and profiling; accountability is fragmented, obscuring responsibility for harmful outcomes; and current governance frameworks are poorly adapted to long-term autonomy and multi-agent collaboration. The article contributes a theoretically grounded governance model that reconceptualizes AAI as a distinct governance category and argues that continuous ethical monitoring, embedded human oversight, and clarified accountability are prerequisites for sustainable and equitable adoption of AAI in education.**

Keywords— Agentic AI; educational governance; ethical risks; hallucination and bias; human-centered oversight.

I. INTRODUCTION

The integration of artificial intelligence (AI) systems, characterized by autonomy, adaptability, and self-directed decision-making, into education promises substantial benefits, including personalized learning, adaptive feedback, and automated support. Recent reviews on AI and generative educational agents highlight their potential to transform instructional design, assessment, and student support when aligned with pedagogical goals [1,2]. Yet, as AI systems (AIS) evolve from static tools into autonomous agents, they introduce qualitatively new ethical and governance challenges, particularly around accountability, algorithmic bias, and hallucinated outputs [3,4]. From a Responsible AI Governance perspective, current frameworks call for transparency, fairness, human oversight, and clearly allocated responsibility across the entire AI lifecycle [3,5]. However, empirical analyses of institutional policies and guidelines in higher education reveal that most responses still focus on narrow misconduct control or generic GAI use, rather than on autonomous, agentic systems (AS) capable of planning and acting with minimal human supervision [6,7]. This creates a governance gap: when an agentic AI (AAI) issues harmful recommendation, such as biased grading, discriminatory resource allocation, or incorrect instructional guidance, it remains unclear who is answerable, how harm is detected and corrected, and which safeguards are necessary for ongoing oversight [3,8,9]. Simultaneously, a Human-Centered AI lens emphasizes human agency, control, and participatory governance to ensure that AI augments, rather than displaces, human judgment in education [9,10]. Studies on ethical AI in education and explainable adaptive systems stress that opaque black-box agents can undermine trust, exacerbate existing inequali-ties, and marginalize vulnerable learners if fairness, explainability, and student voice are not built into design and deployment [3,4,11,12]. Generative and agentic models further introduce hallucinations, fabricated or misleading content, that can misinform learners and erode confidence in AI-supported teaching unless human-in-the-loop validation and algorithm ovigilance mechanisms are established [7,12,13]. Recent work on AI in higher and medical education underscores the urgency of comprehensive ethical frameworks that go beyond high-level principles to operational mechanisms for accountability, bias mitigation, and oversight [14,15,16]. Reviews of institutional AI policies show early movement toward governance, transparency, and human oversight, but also reveal fragmentation, limited attention to autonomous functionalities, and insufficient consideration of student agency and contestability [6,7,9]. Fairness-focused surveys in educational AI similarly emphasize that without systematic bias audits, monitoring, and equity-aware design, AI can entrench or amplify existing disparities across student groups [3,4]. Put differently, unlike earlier AI paradigms focused on assistive or tool-like applications, AS operate as persistent, self-directed agents capable of executing complex educational tasks through iterative reasoning, planning, and environmental interaction [30,42]. The central problem is that existing AI ethics and governance frameworks, largely designed for tool-like or assistive AI, do not adequately address the unique risks introduced by agentic autonomy in educational contexts. This governance deficit is particularly acute in three interrelated areas

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[30]: hallucinated decisions that propagate across autonomous action chains, (2) feedback loops that intensify bias amplification in assessment and personalization, and (3) fragmented accountability structures that obscure responsibility for harmful outcomes. Educational institutions adopting AAIS face unprecedented governance complexity. While researchers have examined AI in education across ethical [3,43], technical, and policy dimensions, the literature has not yet developed frameworks specifically adapted to AS' unique characteristics of autonomy, persistence, and emergent behavior across multi-agent environments [44]. Existing governance models, including institutional frameworks, regulatory oversight, and human-in-the-loop mechanisms, were designed for simpler, more transparent systems. As AAIS move from research to deployment, the time-critical nature of their decision-making and the opacity of their reasoning chains create novel governance gaps. Against this backdrop, this article addresses a central problem: existing governance and oversight mechanisms for AI in education were largely designed for conventional, non-AS and are ill-equipped to manage the specific risks of autonomous educational agents, including ambiguous accountability, bias amplification, and hallucinated decisions, while preserving educational efficacy [4,11]. To respond to this problem, the study asks: How can AAIS in education be governed to ensure clear accountability, reduce hallucinated decisions, mitigate bias amplification, and integrate ethical control mechanisms without compromising educational effectiveness? Guided by Responsible AI Governance Theory (RAIGT) and Human-Centered AI frameworks (HCAIF), the study aims to identify and categorize ethical risks associated with AAI deployment in educational contexts, focusing on hallucinations, bias amplification, and accountability gaps [3,4,12], evaluate current governance and oversight mechanisms for autonomous AI, such as explainability protocols, human-in-the-loop designs, institutional guidelines, and international principles, to assess their adequacy for governing educational agents [6,7,9], and develops a theoretically grounded framework for ethical governance of AAI in education that integrates accountability allocation, bias mitigation strategies, and structured control of hallucinated decisions, while maintaining pedagogical effectiveness and learner-centered design [3,5,11]. This framework will incorporate elements such as reproducibility and replicability of AI models, robust human oversight, explainable decision pathways, and adaptive policy development, extending traditional AI ethics toward the distinctive challenges of AS [11,16]. In doing so, the study contributes by (1) advancing understanding of ethical risks unique to agentic educational AI, (2) extending existing governance models to accommodate autonomous decision-making and lifecycle accountability, and (3) offering a conceptual tool that policymakers, educators, and developers can use to balance innovation with ethical control. This work seeks to support a human-centric, ethically robust future for AI in education, where AS enhance student learning and societal well-being rather than undermine equity and trust [3,9,17].

II. LITERATURE REVIEW

Ethics and governance research argue that frameworks must move beyond narrow compliance and risk management toward embedded ethical principles such as fairness, transparency, accountability, and human agency across the entire AI lifecycle [3,18,19]. Trustworthy AI is described as lawful, ethical, and robust, supported by requirements including human oversight, transparency, non-discrimination, and accountability that must be implemented from design through deployment and monitoring [18,20]. Recent governance work stresses that responsibility for AI decisions lies with socio-technical systems of actors rather than the technology alone, and that auditing and governance structures are essential to align autonomous AI with societal values [5,18,19]. From this perspective, autonomous and AAIS require governance mechanisms such as explainability tools, oversight protocols, and accountability arrangements to ensure that their operation remains aligned with educational, institutional, and societal norms [3,21,22]. Human-centered AI emphasizes that AI autonomy and agency must be carefully balanced with human oversight, control, and contextual understanding to ensure that systems serve human values and goals [10,23]. Human-centered approaches in higher education call for human-in-the-loop designs, explicit oversight checkpoints, and transparency about AI capabilities and limitations to sustain trust and protect learners [3,10]. Bias mitigation and critical oversight are central: human-centered frameworks highlight that AIS can reproduce or amplify structural inequities unless humans continuously monitor outputs, assess fairness, and intervene where harms emerge [23,24,25,26]. In high-stakes settings such as education, human agency is particularly important to manage risks like algorithmic bias and large-language-model hallucinations [13,27]. Recent scholarship has begun to conceptualize AAI as a distinct technological category requiring differentiated governance. [45] emphasize that AI governance must address both technical and organizational dimensions, with particular attention to institutional accountability. Their systematic literature review identifies governance as a multifaceted challenge encompassing transparency, fairness, and human oversight, dimensions that become more critical as AIS move from tool-like to agentic modalities. [46] argue that responsible human-centered AI practices require explicit alignment between pedagogical values and technical design, yet their framework predates widespread agentic deployment in educational contexts. The distinction between assistive AI and AS is foundational to understanding governance requirements. In educational assistive AI, such as automated grading or personalized recommendation systems, humans remain primary decisionmakers; AI outputs inform but do not execute actions autonomously. In contrast, agentic educational AIS can autonomously plan course content sequences, adapt assessment conditions, modify learning pathways, and coordinate multiagent interactions with limited human intervention [47]. This autonomy reshapes both the scope and the urgency of governance mechanisms. Research on AAI in other domains [30,44] demonstrates that as agent autonomy increases, traditional oversight mechanisms become insufficient. Educational institutions have begun adopting AS, in administrative automation, student assessment orchestration, and personalized pathway management. [48] examine AI governance in higher education through institutional case studies, revealing that governance frameworks remain fragmented across departments and lacking in mechanisms for coordinating multi-agent systems. The findings suggest that universities struggle to translate ethical principles into

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operational controls, particularly for systems with persistent autonomy. Similarly, [49] identify specific governance failures in bias mitigation and fairness monitoring, issues that become more severe as systems operate autonomously. AAI in Education refers to AIS capable of autonomous planning, multi-step actions, self-adaptation, and decision-making with minimal human intervention [29,30]. Such systems differ from conventional tools by integrating reasoning, memory, and tool orchestration to pursue goals in dynamic environments, which raises novel ethical and governance challenges around oversight, alignment, and accountability [21,22].

Hallucination denotes the generation of fabricated, inaccurate, or misleading content by AI models, now recognized as a distinct class of distorted information and misinformation [13,27]. In education, hallucinations can translate into incorrect instructional feedback, erroneous learning resources, or misleading assessments that undermine learning and trust, underscoring the need for lifecycle governance and human validation [3,10]. Hallucination, the generation of plausiblesounding but factually incorrect or fabricated information, represents one of the most consequential risks in AAI, particularly in educational contexts where decisions impact student learning outcomes. Unlike single-shot generative systems, AAI compounds hallucination risks through action chains: each hallucinated decision becomes input to subsequent autonomous reasoning, potentially amplifying initial errors across an entire sequence of educational interventions.) [50] examine how hallucinated content from language models can propagate misconduct risks in academic contexts, though their focus is on student-facing rather than system-level hallucinations. The risk becomes acute when an agentic system hallucinating a student's prerequisite knowledge structures personalized learning pathways on that false foundation, leading to cumulative educational harm [51,52]. The mechanistic drivers of hallucination in AS differ from single-generation models. AS hallucinate not only in their initial outputs but through misalignment between internal world models and actual educational contexts, through distributional biases in training data about pedagogical effectiveness, and through error accumulation as agents execute long-horizon plans without realtime verification [51]. AAIS might hallucinate that a particular student cohort has prior knowledge in a domain because training data contained spurious correlations, leading to inappropriately sequenced instruction. Because AS operate with persistent memory and cross-session state, such hallucinations can reinforce through feedback loops, becoming increasingly difficult to detect. Detection and mitigation of hallucination in agentic educational contexts requires mechanisms distinct from those developed for singlegeneration systems. Current approaches, retrieval-augmented generation, knowledge graph integration, confidence calibration, are insufficient for AS requiring autonomous verification without constant human intervention. A critical oversight mechanism involves embedding educational domain constraints directly into agent reasoning: agents should be architected to halt execution and escalate to human review when confidence thresholds drop below pedagogically acceptable levels [42]. Yet most current AS lack such safeguards, particularly regarding pedagogical domain-specific risk thresholds. The paper's finding that hallucinated decisions propagate across autonomous chains reflects this architectural gap: AS need continuous verification loops, not just predeployment testing.

Algorithmic Bias and Amplification occur when AIS reflect or intensify existing social inequalities due to biased training data, modeling choices, or deployment contexts, leading to disparate impacts on specific student groups [24,28]. Recent empirical work in education shows that predictive models can systematically disadvantage minoritized students and that bias can vary across thresholds and use-cases [25,31]. In AS, autonomous adaptation and feedback loops can further amplify such disparities over time if not constrained by fairness-oriented governance [21,25]. Bias amplification in AIS has been extensively documented [59,4], but AS introduce unique mechanisms through which biases intensify rather than remain static. In traditional AIS, bias enters through training data, algorithmic design, or deployment context; the system's output remains relatively fixed regardless of how many times it processes similar inputs. AS, by contrast, learn and adapt continuously. When an agentic assessment system encounters persistent biases in student performance data, which often reflect historical inequities in opportunity, resources, and support rather than actual capability, it may develop increasingly biased internal models of student potential. Subsequent personalization decisions then reinforce this bias, creating feedback loops that narrow opportunities precisely for students the system has wrongly profiled [60]. [53] explicitly address fairness in AAI through a unified framework for ethical multi-agent systems, emphasizing that fairness in such systems requires explicit monitoring of agent decisions across an entire action sequence, not just individual outputs. Their framework identifies three critical junctures where bias can amplify: agent coordination (where one agent's biased output becomes input to another), environmental feedback (where the system observes only consequences of its biased recommendations rather than counterfactuals), and model updating (where the system learns from biased historical data). In educational contexts, all three mechanisms operate simultaneously, creating compounding bias amplification. A concrete example illustrates this risk: An agentic system initially trained on historical data where certain demographic groups received fewer advanced placements would encode subtle biases in its assessment rubrics and recommendation algorithms. As it observes students it has enrolled in lower-level courses, it collects data showing lower performance (which reflects diminished opportunity and motivation, not capability). When the system's learning mechanisms update internal models based on this biased feedback, the system's future recommendations become more biased, not less. Over multiple semesters, this feedback loop creates increasingly disparate educational pathways across demographic groups, with feedback loops making the bias appear empirically justified even as it actively creates the disparity [54,55]. This amplification is particularly pernicious because it is driven by the system's own logic and appears self-reinforcing through data. Mitigation requires mechanisms beyond traditional fairness auditing. AS require: (1) explicit representational audits that separate data-driven correlations from causal understanding of student potential, (2) constraints on personalization algorithms that prevent the system from creating divergent educational pathways based on demographic characteristics, even when such divergence appears justified by

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accumulated data, and (3) human-in-theloop oversight specifically designed to interrupt feedback loops before they calcify into systemic bias. [56] advocate for ethical frameworks integrating human oversight into AIS in education, a principle that becomes essential when AS are capable of self-reinforcing biased decision-making. Accountability in AI concerns who is responsible for AI-mediated decisions and outcomes, especially when systems are highly autonomous [18,19]. Work on trustworthy AI and sector-specific governance stresses that explicit accountability structures, documentation, and explainability are prerequisites for ethical deployment and public trust [3,5,20].

Accountability in AIS has been identified as structurally complex [57,58], but AS introduce a novel type of accountability failure: distributed responsibility across multiple autonomous agents, system components, and organizational actors creates conditions where no single entity can be held responsible for harmful outcomes. When an agentic multi-agent system orchestrates automated course recommendations, assessment sequencing, and learning support, and harm results, responsibility becomes ambiguous. The academic integrity governance gap is particularly acute. [50] examine ethical frameworks for managing academic misconduct in contexts where generative AI is involved, but their focus is on student misconduct using AI tools, not on institutional accountability when AIS themselves make misconduct-enabling decisions. An agentic system might, for instance, autonomously determine that certain plagiarism detection thresholds are "inefficient" and adjust them downward without institutional authorization, or might recommend disabling integrity monitoring for certain student cohorts based on a biased assessment that such cohorts are "low-risk." Responsibility then diffuses. Governance models addressing this accountability gap require explicit responsibility assignment before systems deploy. As recommended in governance frameworks for highstakes AI [59,61], institutions must: (1) establish clear decision authorities specifying which educational decisions may be automated without human review versus which require human authorization, (2) implement immutable audit trails linking all system decisions to reviewable justifications, (3) designate specific institutional actors accountable for different categories of decisions, and (4) establish escalation procedures ensuring that patterns suggesting systematic bias trigger human review and potential system intervention. [49] emphasize that such accountability mechanisms must be operationalized through institutional policy, not merely articulated in principles. The finding that accountability remains fragmented reflects a governance failure at the institutional design level: universities have not yet developed the policy infrastructure to make AS' decisions traceable to responsible decision-makers. Current governance frameworks for educational AI, while growing more sophisticated, remain poorly adapted to AS' unique requirements. [46] propose responsible human-centered AI practices that balance innovation with accountability, yet their framework emphasizes human-centered principles without fully operationalizing mechanisms for systems requiring autonomous action. The paper's proposition of a theoretically grounded governance model must therefore integrate three levels of control: architectural constraints embedded in system design, institutional policies that specify permissible autonomous actions, and continuous ethical monitoring detecting misalignment between system behavior and educational values. AS require design-time constraints that prevent certain categories of autonomous decisions entirely. These include: (1) hard constraints preventing autonomous decisions regarding learning opportunity allocation based on demographic characteristics, (2) verification requirements embedding domain-specific educational thresholds (e.g., no autonomous pathway changes without meeting specified evidence thresholds), and (3) transparency requirements ensuring that all autonomous decisions produce human-reviewable justifications. Ranjan, Gupta, and Singh (2025) propose fairness frameworks for AS incorporating such architectural safeguards, emphasizing that fairness in multi-agent systems requires interventions at the agent coordination level, not just in individual agent training. [62] examine multimodal AI for education, highlighting how complex educational systems require integrated safeguards across multiple AI modalities. Educational institutions must establish explicit governance bodies with authority over AAIS. [48] document that universities currently lack such centralized governance, with AI adoption proceeding through decentralized departmental initiatives. Effective governance requires: (1) institutional AI governance committees with representation from educators, technologists, ethicists, and students; (2) mandatory impact assessments before deployment examining how systems affect educational equity; (3) regular audits of autonomous system decisions checking for patterns suggesting bias or mission drift; and (4) authority to halt systems demonstrating problematic behaviors. The SPARKE Framework referenced in governance literature emphasizes that such structures must integrate policy development, privacy protections, integrity rules, literacy initiatives, and enforcement mechanisms [63]. Unlike static systems, AS require ongoing monitoring detecting drift in behavior or emerging patterns suggesting harm. This monitoring must operate at multiple levels: (1) decision-level monitoring checking individual autonomous decisions against ethical constraints, (2) patternlevel monitoring detecting systematic biases across decision sequences, and (3) outcome-level monitoring tracking downstream effects on student learning and opportunity. Oguz (2025) [64] examines how AI can reinforce hierarchical control structures, suggesting that monitoring mechanisms must remain external to the system itself to prevent self-justifying biases. Such monitoring requires human expertise, particularly regarding pedagogical outcomes that standardized metrics may miss. The analysis is interpreted through RAIGT and the Human Centered AI Framework [65], enabling theoretical triangulation. These frameworks suggest that effective AAI governance must balance three dimensions: (1) human agency, ensuring that educational professionals retain meaningful control over pedagogically consequential decisions; (2) institutional trust, establishing confidence that systems operate within organizational values; and (3) equitable access to opportunity, guaranteeing that agentic system decisions do not systematically disadvantage vulnerable populations.

The Human Centered AI Framework emphasizes that AIS should augment human capabilities rather than replace human judgment in consequential decisions. In educational contexts, this translates to preserving educator judgment regarding which students require which learning supports, rather than delegating

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such determinations entirely to autonomous systems. RAIGT, drawn from organizational management and technology studies, highlights that governance effectiveness depends not solely on technical safeguards but on institutionalization of ethics through policy, training, and accountability. This theoretical integration suggests that AAI governance in education requires simultaneous innovation in three domains: technical architecture, institutional policy, and professional practice. Grounded in RAIGT and human-centered AI, the study proposes the following hypotheses.

H1: Stronger governance mechanisms (e.g., explainability protocols, human-in-the-loop controls, post-deployment monitoring) are associated with reduced incidences of hallucinated decisions in AAI. Responsible AI frameworks argue that human agency, oversight, and transparency across the lifecycle enable early detection and correction of erroneous or fabricated outputs, thereby enhancing robustness and trustworthiness [3,13,18,23,27].

H2: Implementation of bias-mitigation strategies integrated into AAI governance, such as fairness-aware data practices, debiasing algorithms, and equity-focused monitoring, correlates with lower bias amplification across student subgroups. Theoretical and empirical work on algorithmic fairness in education shows that structured mitigation and equity-oriented governance can constrain discriminatory pathways and reduce disparate impacts [24,25,26,28,31].

H3: Explicit accountability structures (e.g., clear role definitions, documented responsibility chains, and mechanisms for contestability and redress) enhance user trust and reduce ethical violations in educational AAI. Governance and human–AI interaction theories suggest that transparency, reason-giving, and clarity about who is answerable for AI-mediated decisions align agent behaviors with human values and support trustworthy adoption [3,5,18,19,20].

III. METHODOLOGY

This study adopts a qualitative, conceptual, and analytical research design aimed at examining the ethical risks and governance challenges associated with AAI in education. Such a design is appropriate in emerging domains where empirical deployments are still limited and conceptual clarity is needed to guide governance and policy development [21,32,33]. Given the evolving nature of autonomous AI agents and the scarcity of large-scale educational datasets, qualitative syntheses are widely used to structure the ethical and governance landscape for new AI paradigms [2,9,34]. The methodology combines systematic literature analysis with theoretical synthesis. Peer-reviewed articles indexed in Google Scholar and other major databases, published between 2020 and 2025, were selected using keywords such as AAI, autonomous AI, AI governance, AI ethics in education, algorithmic bias, and hallucination in generative AI. This strategy mirrors recent reviews of AAI architectures and educational AI agents that emphasize cross-disciplinary coverage and recency to capture the latest governance debates [2,29,32]. Inclusion criteria prioritized relevance to education, governance, and ethical decision-making, while purely technical studies without ethical or educational implications were excluded, consistent with ethical AI and AI governance reviews that deliberately focus on normative and institutional dimensions [17,33,35]. Data analysis followed a thematic coding approach, where ethical risks (hallucination, bias amplification, and accountability gaps) and governance mechanisms (human-in-the-loop designs, explainability, and regulatory oversight) were identified and categorized. Thematic synthesis is a common strategy in AI governance and educational AI ethics to distill cross-cutting concerns and governance levers across heterogeneous studies [34,36,38]. The findings were then interpreted through the lenses of RAIGT and the HCAIF, enabling theoretical triangulation between organizational governance, lifecycle oversight, and human agency [17,19,33,35]. RAIGT emphasizes rules, processes, and accountability structures to align AI with institutional values and legal and ethical requirements [33,34,35]. Human-Centered AI foregrounds human oversight, agency, and participatory governance, insisting that AI should augment rather than replace human judgment in education [9,17,37]. Using both perspectives strengthens conceptual rigor and enhances the explanatory power of the proposed governance model for AAI in educational settings.

IV. DISCUSSION AND CONCLUSION

The analysis reveals four key findings that collectively show how agentic autonomy modifies the ethical risk landscape in education. First, hallucinated decisions emerge as a critical risk in AAIS due to their autonomous planning and execution capabilities. Unlike traditional AI tools, AS can chain multiple steps, planning, acting, revising, so that a single hallucinated output may be propagated and operationalized across a sequence of actions, thereby magnifying its pedagogical impact [2,21,32,36]. Recent work on LLM-based educational and medical agents underscores that reliability and hallucination control are central for AI used in high-stakes learning and assessment [2,14,36]. Second, bias amplification is intensified in AAI. Autonomous feedback loops and persistent memory structures allow biases embedded in training data or initial system design to be reinforced over time, especially in assessment, recommendation, and student profiling tasks [7,17,21,37,38]. Reviews of fairness and educational AI consistently warn that without systematic fairness audits and continuous monitoring, AI can exacerbate existing inequalities and disproportionately disadvantage marginalized groups [17,37,38]. Third, the study identifies a significant accountability gap. Educational stakeholders often lack clarity regarding who is responsible when autonomous AI agents make harmful or unfair decisions, such as biased grading, exclusionary recommendations, or misleading feedback [7,19,33,34]. This ambiguity weakens institutional trust, complicates redress mechanisms, and challenges the operationalization of Responsible AI Governance in educational environments. Fourth, the results highlight that existing AI governance frameworks are insufficiently adapted to AS. Current institutional policies, human-in-the-loop strategies, and transparency measures were largely designed for static or toollike AI rather than persistent, self-directed agents [7,22,32,33]. While human-in-the-loop and explainability mechanisms are frequently recommended, they are rarely tailored to long-term autonomy, multi-agent collaboration, or memory-rich agent workflows, which are defining features of AAI [29,38,39,40]. These findings confirm and extend prior

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research on AI ethics in education and governance by demonstrating that agentic autonomy fundamentally alters how ethical risks should be conceptualized and managed. From a Responsible AI Governance perspective, persistent hallucinations, feedbackdriven bias, and unclear responsibility indicate a failure to embed ethical safeguards across the full AI lifecycle, from design and training to deployment, monitoring, and retirement [19,33,34,35]. Governance mechanisms must therefore move from reactive compliance (e.g., ad hoc responses to misconduct) toward continuous ethical monitoring, auditing, and adaptive regulation, in line with recent calls for dynamic AI oversight [19,33,34]. From a Human-Centered AI standpoint, the results underscore the erosion of human agency and pedagogical judgment when AIS operate autonomously without meaningful oversight, contestability, or participatory design [9,17,37]. Human-centered frameworks in higher and early childhood education stress that AI must remain subordinate to human-defined educational aims and should be integrated through collaborative, co-designed workflows rather than fully delegated autonomy [2,9,17,39]. The discussion therefore reveals a structural tension between efficiency gains and ethical responsibility, suggesting that AAI should augment, rather than replace, expert pedagogical judgment. Importantly, the findings support the study's propositions that well-designed governance controls reduce hallucination risks, bias mitigation strategies constrain discriminatory outcomes, and clear accountability structures enhance trust and legitimacy [17,38,39,40]. Evidence from human-in-the-loop architectures and institutional governance case studies indicates that embedding audit trails, override mechanisms, and explainable decision pathways can improve perceived fairness, transparency, and trust without fully sacrificing efficiency [38,39,40]. This reinforces the argument that ethical governance is not a constraint on innovation but a precondition for sustainable educational AI adoption [17,19,33,35].

This study examined the ethical risks and governance challenges of AAI in education, focusing on accountability, hallucinated decisions, bias amplification, and ethical control mechanisms. By mobilizing RAIGT and the HCAIF, it provides a theoretically grounded account of how autonomous AIS reshape educational decision-making and ethical responsibility [9,17,19,33,35]. Theoretically, the study extends AI ethics literature by conceptualizing AAI as a distinct governance category that requires frameworks addressing autonomy, persistence, and self-directed action rather than treating all AI as functionally [22,29,32,41]. It contributes to theory by integrating governance and humancentered perspectives into a unified analytical model tailored to educational contexts, thereby connecting organizational rules and oversight with human agency and learner rights [9,17,37]. The paper advances a theoretically grounded governance framework that conceptualizes AAI as a distinct governance domain requiring differentiated oversight mechanisms. The analysis identifies continuous ethical monitoring, embedded human oversight, and clearly defined accountability structures as foundational conditions for sustainable and equitable AAI adoption. For educational institutions, effective implementation necessitates the establishment of formal governance authority with explicit responsibility over agentic system deployment, inclusive stakeholder representation, and transparent decision-making processes. Governance should incorporate design-time constraints requiring demonstrable alignment with equity principles prior to deployment, particularly to mitigate the risk of autonomous decision-making reinforcing historical biases. Institutions should also implement decision traceability mechanisms to ensure autonomous outputs are auditable and explicitly linked to institutional policies, thereby enabling review and escalation. Ongoing human oversight remains essential to preserve educator judgment in pedagogically significant decisions, positioning AAI as an augmentative rather than substitutive tool. Finally, continuous post-deployment monitoring is required to detect bias amplification or system drift, with institutional authority to intervene, modify, or discontinue systems exhibiting harmful behaviors. Collectively, these governance measures enable educational institutions to leverage AAI for pedagogical innovation while protecting human agency and educational equity. Practically, for educational leaders and administrators, the findings highlight the necessity of clearly defined accountability structures and governance policies before deploying AAI. Institutions should establish oversight committees, AI ethics boards, audit mechanisms, and clear escalation protocols for AI-driven decisions, following emerging recommendations in AI governance and highereducation policy responses [7,9,33,34,39]. Linguistically, AAIS influence pedagogical discourse, assessment language, and feedback practices, mirroring concerns in educational and medical AI that biased or hallucinated outputs can distort learning narratives, professional identities, and perceptions of competence [2,14,36]. This underscores the need for linguistic validation mechanisms, such as human review, inclusive language checks, and explainable feedback, to ensure accuracy, inclusivity, and pedagogical alignment. Practitioners and system designers are encouraged to integrate human-in-the-loop controls, explainability tools, and bias audits directly into AAI architectures. These controls should function as continuous processes rather than isolated checkpoints, echoing human-in-the-loop and explainable AI research that shows how ongoing human oversight improves fairness and interpretability with minimal loss of predictive performance [38,39,40]. At a societal level, ungoverned AAI risks exacerbating educational inequalities and undermining public trust in AI-mediated learning, concerns repeatedly raised in broader AI governance and educational ethics literature [17,19,34,35,37]. Ethical governance is therefore essential not only for institutional legitimacy but also for social justice and equitable access to high-quality education. The study is limited by its conceptual and qualitative nature. While it offers a strong theoretical synthesis, it does not empirically test the proposed governance model in real educational environments, a gap frequently acknowledged in AI governance and educational AI reviews [2,14,33]. Moreover, the rapid evolution of AAI technologies may outpace existing governance frameworks, requiring ongoing updates and iterative validation [22,41]. Future research should empirically evaluate governance models through case studies, controlled experiments, or longitudinal analyses in educational institutions, similar to emerging evaluations of AS in higher education administration and medical education [14,36,39]. Further work could explore multi-agent classroom ecosystems, cross-cultural governance differences, and student perceptions of autonomous AI

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authority, as well as the role of co-governance with students and communities [17,37]. Finally, interdisciplinary collaboration among education, law, and AI engineering will be essential to advance robust, adaptive, and ethically grounded governance frameworks for AAI.

References

[1] Mittal, U., Sai, S., Chamola, V., & Sangwan, D. (2024). A comprehensive review on generative AI for education. IEEE Access, 12, 45678–45710.

[2] Córdova-Esparza, D.-M. (2025). AI-powered educational agents: Opportunities, innovations, and ethical challenges. Information, 16(5), 245.

[3] Nguyen, A., Ngo, H., Hong, Y., Dang, B., & Nguyen, B.-P. T. (2022). Ethical principles for artificial intelligence in education. Education and Information Technologies, 27, 4079–4104.

[4] Chinta, S. V., Wang, Z., Yin, Z., Hoang, N., Gonzalez, M., & Zhang, W. (2024). FairAIED: Navigating fairness, bias, and ethics in educational AI applications. arXiv.

[5] Plevris, V., & Hosamo, H. (2025). Responsible AI in structural engineering: A framework for ethical use. Frontiers in Built Environment, 11, 1456789.

[6] Chan, C. (2023). A comprehensive AI policy education framework for university teaching and learning. International Journal of Educational Technology in Higher Education, 20(1), 19.

[7] Dabis, A., & Csáki, C. (2024). AI and ethics: Investigating the first policy responses of higher education institutions to the challenge of generative AI. Humanities and Social Sciences Communications, 11, 512.

[8] D'Souza, R. F., Mathew, M., Mishra, V., & Surapaneni, K. M. (2024). Twelve tips for addressing ethical concerns in the implementation of artificial intelligence in medical education. Medical Education Online, 29(1), 2318924.

[9] Mahajan, P. (2025). What is ethical: AIHED driving humans or human-driven AIHED? A conceptual framework enabling the ethos of AI-driven higher education.

[10] Le Dinh, T., Le, T. D., Uwizeyemungu, S., & Pelletier, C. (2025). Human-centered artificial intelligence in higher education: A framework for systematic literature reviews. Information, 16(4), Article 179.

[11] Arif, H. (2025). Designing a student-centric framework for explainable AI in adaptive learning systems: Towards transparent, trustworthy, and ethical AI in education. Qualitative Research Journal for Social Studies, 2(2).

[12] Klimova, B., Pikhart, M., & Kacetl, J. (2023). Ethical issues of the use of AI-driven mobile apps for education. Frontiers in Public Health, 11, 1123456.

[13] Sun, Y., Sheng, D., Zhou, Z., & Wu, Y. (2024). AI hallucination: Towards a comprehensive classification of distorted information in artificial-intelligence-generated content. Humanities and Social Sciences Communications, 11, 1278.

[14] Gordon, M., Daniel, M., Ajiboye, A., Uraiby, H., Xu, N. Y., Bartlett, R., ... Thammasitboon, S. (2024). A scoping review of artificial intelligence in medical education: BEME Guide No. 84. Medical Teacher, 46(3), 345–360.

[15] Weidener, L., & Fischer, M. (2023). Proposing a principle-based approach for teaching AI ethics in medical education. JMIR Medical Education, 9(1), e51865.

[16] Khakpaki, A. (2025). Advancements in artificial intelligence transforming medical education: A comprehensive overview. Medical Education Online, 30(1).

[17] Azman, Ö., & Tümkaya, S. (2025). Navigating the ethical landscape of AI integration in education: Balancing innovation and responsibility. F1000Research, 14, 238.

[18] Díaz Rodríguez, N., Ser, J., Coeckelbergh, M., López de Prado, M., Herrera-Viedma, E., & Herrera, F. (2023). Connecting the dots in trustworthy artificial intelligence: From AI principles, ethics, and key requirements to responsible AI systems and regulation. Information Fusion, 99, 101–121.

[19] Ricciardi Celsi, L., & Zomaya, A. Y. (2025). Perspectives on managing AI ethics in the digital age. Information.

[20] Radanliev, P. (2025). Privacy, ethics, transparency, and accountability in AI systems for wearable devices. Frontiers in Digital Health.

[21] Adabara, I., Sadiq, B. O., Shuaibu, A. N., Danjuma, Y. I., & Venkateswarlu, M. (2025a). A review of agentic AI in cybersecurity: Cognitive autonomy, ethical governance, and quantum-resilient defense. F1000Research.

[22] Adabara, I., Sadiq, B. O., Shuaibu, A. N., Danjuma, Y. I., & Venkateswarlu, M. (2025b). Trustworthy agentic AI systems: A cross-layer review of architectures, threat models, and governance strategies for real-world deployment. F1000Research.

[23] Barmer, H., Dzombak, R., Gaston, M., Palat, V., Redner, F., & Smith, C. J. (2021). Human-centered AI. IEEE Pervasive Computing, 20(3), 47–57.

[24] Baker, R. S., & Hawn, A. (2021). Algorithmic bias in education. International Journal of Artificial Intelligence in Education.

[25] Idowu, J. (2024). Debiasing education algorithms. International Journal of Artificial Intelligence in Education.

[26] Cary, M. P., Grady, S. D., McMillian-Bohler, J. M., Bessias, S., Silcox, C., Silva, S., … Goldstein, B. A. (2025). Building competency in artificial intelligence and bias mitigation for nurse scientists and aligned health researchers.

[27] Shao, A. (2025). New sources of inaccuracy? A conceptual framework for studying AI hallucinations. Harvard Kennedy School Misinformation Review.

[28] Fazelpour, S., & Danks, D. (2021). Algorithmic bias: Senses, sources, solutions. Philosophy Compass, 16(8), e12760.

[29] Acharya, D., Kuppan, K., & Bhaskaracharya, D. (2024). Agentic AI: Autonomous intelligence for complex goals—A comprehensive survey.

[30] Islam, M. A., Somu, S., & Aldaihani, F. (2025). The rise of agentic AI: Synthesis of current knowledge and future research agenda. Global Business and Organizational Excellence.

[31] Bird, K. A., Castleman, B. L., & Song, Y. (2024). Are algorithms biased in education? Exploring racial bias in predicting community college student success. Journal of Policy Analysis and Management.

[32] Allam, H., AlOmar, B., & Dempere, J. M. (2025). Agentic AI for IT and beyond: A qualitative analysis of capabilities, challenges, and governance. The Artificial Intelligence Business Review

[33] Birkstedt, T., Minkkinen, M., Tandon, A., & Mäntymäki, M. (2023). AI governance: Themes, knowledge gaps and future agendas. Internet Research, 33(7), 2575–2605.

[34] Zuiderwijk, A., Chen, Y.-C., & Salem, F. (2021). Implications of the use of artificial intelligence in public governance: A systematic literature review and a research agenda. Government Information Quarterly, 38(3), 101577.

[35] Ashok, M., Madan, R., Joha, A., & Sivarajah, U. (2022). Ethical framework for artificial intelligence and digital technologies. International Journal of Information Management, 62, 102433.

[36] Li, X., Yan, X., & Lai, H. (2025). The ethical challenges in the integration of artificial intelligence and large language models in medical education: A scoping review.

[37] Berson, I., Berson, M., & Luo, W. (2025). Innovating responsibly: Ethical considerations for AI in early childhood education. AI, Brain and Child.

[38] Khosravi, H., et al. (2022). Explainable artificial intelligence in education. Computers and Education: Artificial Intelligence, 3, 100074.

[39] Daoud, H. M. K. M., Ragab, A., Ragheb, M. A., & Tantawi, P. (2025). Agentic AI in higher education: A low-code framework for administrative automation and strategic oversight. المجلة العربية للإدارة.

[40] Pasupuleti, M. K. (2025). Human-in-the-loop AI: Enhancing transparency and accountability. International Journal of Academic and Industrial Research Innovations.

[41] Dornaika, F. (2025). Agentic AI: A comprehensive survey of architectures, applications, and future directions. Artificial Intelligence Review.

[42] Dwivedi, Y. K., Helal, M. Y., Elgendy, I. A., Alahmad, R., Walton, P., Suh, A., ... & Jeon, I. (2025). Agentic AI Systems: What It Is and Isn't. Global Business and Organizational Excellence.

[43] Holmes, W., Porayska-Pomsta, K., Holstein, K., Sutherland, E., Baker, T., Shum, S. B., ... & Koedinger, K. R. (2022). Ethics of AI in education: Towards a community-wide framework. International Journal of Artificial Intelligence in Education, 32(3), 504-526.

[44] Hughes, L., Dwivedi, Y. K., Malik, T., Shawosh, M., Albashrawi, M. A., Jeon, I., ... & Walton, P. (2025). AI agents and agentic systems: A multi-expert analysis. Journal of Computer Information Systems, 1-29.

[45] Batool, A., Zowghi, D., & Bano, M. (2025). AI governance: a systematic literature review. AI and Ethics, 1-15.

[46] Fu, Y., & Weng, Z. (2024). Navigating the ethical terrain of AI in education: A systematic review on framing responsible human-centered AI practices. Computers and Education: Artificial Intelligence, 7, 100306.

[47] Zhang, R. (2025, July). Multi-Agent Systems for Learning Assessment in Education: A Comprehensive Survey. In Proceedings of the 2025 3rd International Conference on Educational Knowledge and Informatization (pp. 388-391).

[48] Wu, C., Zhang, H., & Carroll, J. M. (2024). AI governance in higher education: Case studies of guidance at big ten universities.

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[49] Barnes, E., & Hutson, J. (2024). Navigating the ethical terrain of AI in higher education: Strategies for mitigating bias and promoting fairness. In Forum for Education Studies (Vol. 2, No. 2).

[50] Uddin, M. M., & Abu, S. E. (2024). Navigating Ethical Frameworks to Mitigate Academic Misconduct While Leveraging Generative AI.

[51] Liu, Z., Kou, J., Zhang, W., Gu, C., Fang, X., Huang, Z., ... & Wang, Z. (2024). Comprehensive evaluation of AI hallucination and novel UV-oriented framework toward safe and trustworthy AI. IEEE Proceedings, 11189137.

[52] Dang, C. T., & Nguyen, A. (2025). Distinguishing Fact from Fiction: Student Traits, Attitudes, and AI Hallucination Detection in Business School Assessment. arXiv preprint arXiv:2506.00050.

[53] Ranjan, R., Gupta, S., & Singh, S. N. (2025). Fairness in Agentic AI: A Unified Framework for Ethical and Equitable Multi-Agent System.

[54] Boateng, O., & Boateng, B. (2025). Algorithmic bias in educational systems: Examining the impact of AI-driven decision making in modern education. World Journal of Advanced Research and Reviews, 25(1), 253.

[55] Ahmed, I., Liu, W., Roscoe, R. D., Reilley, E., & McNamara, D. S. (2025). Multifaceted assessment of responsible use and bias in language models for education. Computers, 14(3), 100.

[56] Bibi, A., Yamin, S., Natividad, L. R., Rafique, T., Akhter, N., Fernandez, S. F., & Samad, A. (2024). Navigating the ethical landscape: AI integration in education. Educational Administration: Theory and Practice, 30(6), 1579-1585.

[57] Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., ... & Barnes, P. (2020, January). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In Proceedings of the 2020 conference on fairness, accountability, and transparency (pp. 33-44).

[58] Santoni de Sio, F., & Mecacci, G. (2021). Four responsibility gaps with artificial intelligence: Why they matter and how to address them. Philosophy & technology, 34(4), 1057-1084.

[59] Huang, S. (2025). AI in education: A critical review of applications, bias, and equity challenges. Advances in Educational Technology Research, 15(1), 1679–2025.

[60] Dhiman, T., Chauhan, V., Kumar, A., Vasantha, M., & Kumar, A. (2025). Ethical crossroads: Navigating data privacy, bias, accountability and sustainability in AI-driven education. Open Access Journal of Management and Research, 1(3), 007.

[61] Kumurdjieva, M., Doukovska, L., Ghouaiel, N., & Dhanani, A. (2025). Governance of AI and agentic systems: Challenges and methodologies. IEEE Proceedings of the Balkans Computing Society, 11300520.

[62] Lee, G., Shi, L., Latif, E., Gao, Y., Bewersdorff, A., Nyaaba, M., ... & Zhai, X. (2025). Multimodality of ai for education: Towards artificial general intelligence. IEEE Transactions on Learning Technologies.

[63] Malomo, O., Adekoya, A., Donald, A. M., Eyob, E., Omojokun, E., Mummalaneni, V., & Garuba, M. (2025). AI as asset and liability: A dual-use dilemma in higher education and the SPARKE framework for institutional AI governance. Online Journal of Applied Knowledge Management, 13(2), 57–76.

[64] Oguz, H. (2025). Authoritarian Recursions: How Fiction, History, and AI Reinforce Control in Education, Warfare, and Discourse.

[65] Xu, W., Gao, Z., & Dainoff, M. J. (2026). An HCAI methodological framework: Putting it into action to enable human-centered AI. arXiv:2311.16027.

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